

Total no. of Pages : 01
 Roll. No.:
 IInd SEMESTER, B. TECH (ECAM 1&2)
 MID-SEMESTER EXAMINATION, February, 2025
 Course Code: EAEPC202
 Course Title: Electronic devices and circuits
 Time: 1:30 Hours
 Max. Marks: 20
 Note: Attempt all questions.

Q.no	Question	Marks	CO
1(a)	The intrinsic resistivity of Ge is $0.45 \Omega\text{-m}$. The electron and hole mobilities at room temperature are 0.39 and $0.19 \text{ m}^2/\text{V}\cdot\text{s}$ respectively. Calculate electrons and holes density. Also find electron and hole drift currents for a field of 10 KV/m .	2	CO1
1(b)	Consider an n-type Si for which the donor concentration $N_D = 10^{15}/\text{cm}^3$. Calculate the electron and hole concentration at room temperature. (Assume $n_i = 1.5 \times 10^{10}/\text{cm}^3$)	2	CO1
2(a)	Over what range of input voltage will the Zener circuit shown below maintain 30 V across 2000Ω load, assuming that series resistance $R = 200 \Omega$ and Zener current rating is 25 mA ?	2	CO1
2(b)	Explain the physical structure and operation of Varactor diode with a suitable formula.	2	CO1
3(a)	Explain the working of the circuits for the input waveform	2	CO2

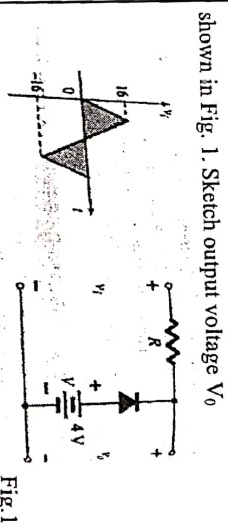
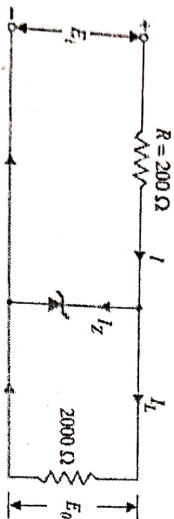


Fig.1



Fig.2

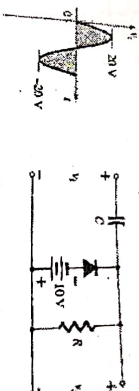


Fig.3

3(b)	Draw the centre-tap full wave rectifier with load $R_L = 1 \text{ k}\Omega$. Each diode has dynamics resistance r_d of 10Ω . The voltage across the secondary winding is $220 \sin 314t$. Find (i) peak value of current (b) dc average value of current (c) rms value of current.	2	CO2
4(a)	Sketch the energy band diagram of pn-junction. At a temperature of 27°C , determine the thermal voltage	2	CO2
4(b)	Determine I_D , V_{D2} and V_o for the circuit shown in Fig.2	2	CO2
5(a)	Explain Drift and Diffusion in pn junction, derive an expression for total hole and electron currents.	2	CO1
5(b)	Explain the working of the circuit with input waveform shown in. Fig.3. Sketch the output waveform.	2	CO2